ARTICLE IN PRESS

Marine Pollution Bulletin xxx (xxxx) xxx–xxx

Contents lists available at ScienceDirect

Marine Pollution Bulletin

journal homepage: www.elsevier.com/locate/marpolbul

Effects of hydrocarbon pollution in the structure of macrobenthic assemblages from two large estuaries in Brazil

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ARTICLE INFO

Keywords:
PAH
Macrofauna
Spatial variation
Multivariate analysis
Paranaguá Bay
Guanabara Bay

ABSTRACT

Changes in the structure of benthic macrofauna and its relationship with hydrocarbon contamination were determined at different spatial scales in sublittoral sediments of two large estuaries in Brazil. Guanabara Bay (GB) is a heavily polluted estuary due to the presence of a large industrial complex and high demographic density. Laranjeiras Bay (LB) lies in an Environmental Protection Area and can still be considered as preserved from human activities. Despite some spatial differences within each bay, the PAHs concentrations were significantly and consistently higher in GB, with values generally above the threshold effect levels. No signs of hydrocarbon contamination were observed in LB. Macrofauna abundance, diversity and overall assemblage structure were largely different between bays. Canonical analysis of principal coordinates (CAP), used to model the relationship between macrofauna and PAHs levels, indicated that this class of hydrocarbons is the main structuring factor of soft-bottom assemblages in both bays.

1. Introduction

Estuaries and shallow bays act as sinks for sediment and associated particle-reactive contaminants (Wang et al., 2012) and, therefore, are among the coastal ecosystems most threatened by anthropogenic activities (McLuski and Elliot, 2006). Aliphatic and polycyclic aromatic hydrocarbons are ubiquitous contaminants in estuaries, particularly those characterized by high urban and industrial development (Colombo et al., 2005; Chen et al., 2013; Dudhagara et al., 2016).

Aliphatic hydrocarbons (AHs) have several sources, which include biogenic and man-induced inputs. Although they may be synthesized by marine organisms, higher plants, bacteria, phytoplankton and zooplankton, AHs are also part of petroleum-related products (Wang et al., 2009). On the other hand, polycyclic aromatic hydrocarbons (PAHs) are predominantly derived from anthropogenic sources, including the incomplete combustion of fossil fuels, coal and plant biomass, in addition to crude oil and its derivatives (Liu et al., 2009).

Due to their hydrophobic characteristics, oil-derived hydrocarbons tend to adsorb on to particulate material and settle to sediments, where they may affect important ecosystem functions such as decomposition rates, oxygen dynamics and nutrient recycling (Law and Biscaya, 1994; Venturini et al., 2008; Cibic et al., 2012). Petroleum by-products are also known to cause adverse effects at different levels of biological organization, from antioxidant defense responses and cellular damage (Morales-Caselles et al., 2008; Sureda et al., 2011; Marques et al., 2014; Sandrini-Neto et al., 2016) to changes in assemblage structure over large spatial scales (Andersen et al., 2008; Ocon et al., 2008; Yu et al., 2013). Soft-bottom macroinvertebrates are frequently used as indicators of pollution because they form abundant and diverse assemblages of species that exhibit different tolerances to stress (Dauvin et al., 2010). Moreover, benthic organisms are relatively sedentary and live in close association with sediments, where contaminants tend to accumulate (Hyland et al., 2005).

Guanabara Bay (GB) is a large estuary located in the Rio de Janeiro metropolitan region (southeastern Brazil) widely known for its high pollution degree (e.g., Carreira et al., 2002; Wagener et al., 2012; Soares-Gomes et al., 2016). GB is an example of a tropical system under long and severe environmental pressure, with practically all its extension showing relatively high concentrations of petroleum-related hydrocarbons (Wagener et al., 2012). On the other hand, Laranjeiras Bay (LB) lies in an extensive Environmental Protection Area on the coast of Paraná state (south Brazil). In general, concentrations of aliphatic and aromatic hydrocarbons in LB sediments are below the threshold effect levels used in environmental monitoring (Martins et al., 2012).

In this work, we evaluated changes in the concentration of hydrocarbons and the structure of benthic macrofauna at different spatial

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http://dx.doi.org/10.1016/j.marpolbul.2017.07.074
Received 6 March 2017; Received in revised form 16 June 2017; Accepted 31 July 2017
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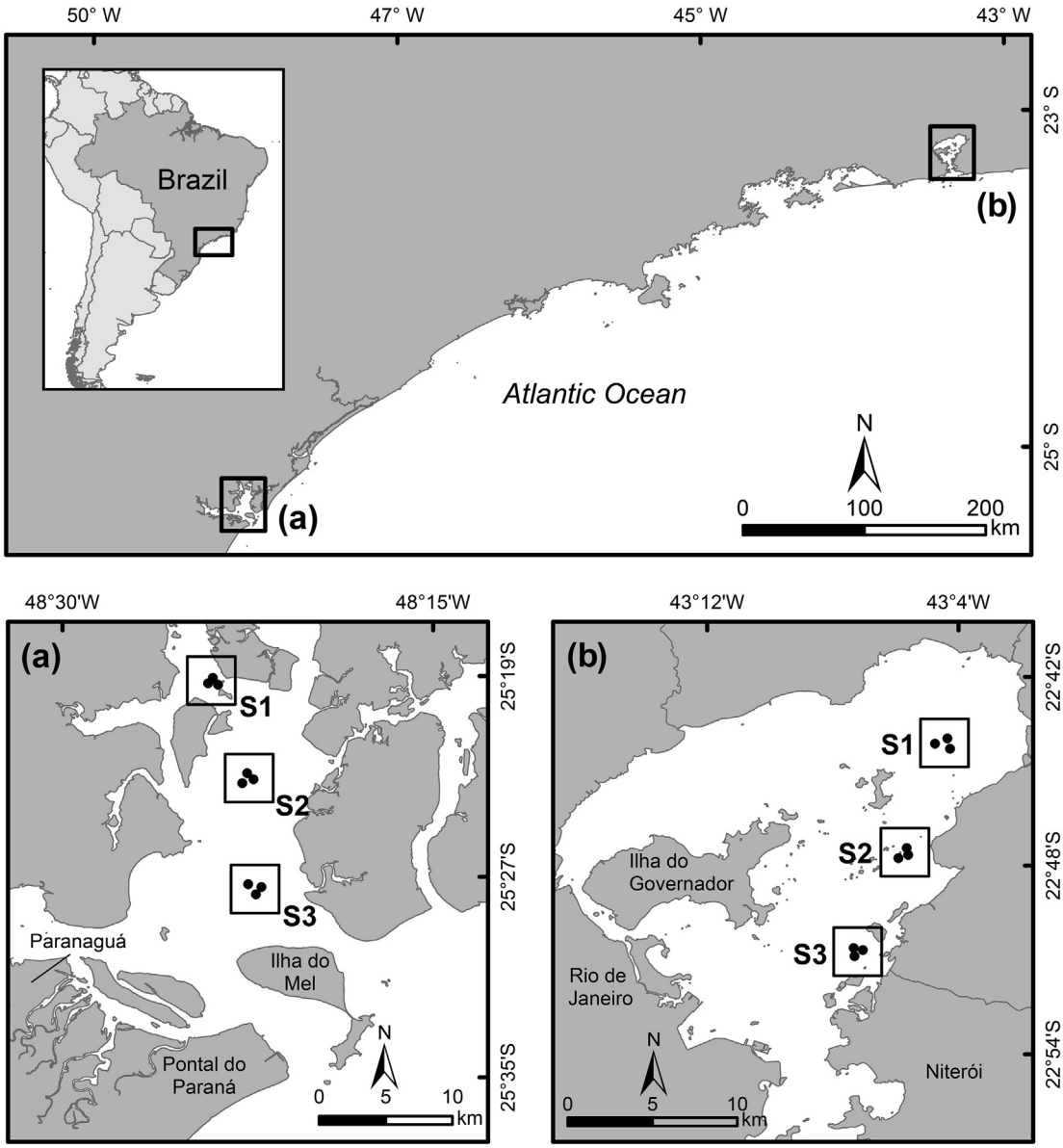

Fig. 1. Map of the study area showing sectors and locations within (a) Laranjeiras Bay and (b) Guanabara Bay. S1 = inner sector; S2 = intermediate sector; S3 = external sector.

scales in sublittoral sediments of GB and LB using a hierarchical sampling design. The incorporation of multiple sources of spatial variation is crucial to detect and measure human impacts on benthic assemblages (Underwood, 2000), mainly in estuaries, which are characterized by high environmental variability (Dauvin and Ruellet, 2009). Moreover, we modelled the relationship between soft-sediment benthic fauna and the concentration of PAHs in sediments of both bays, using canonical analysis of principal coordinates according to Anderson (2008). We hypothesized that the largest proportion of total variation in macrofauna density, diversity and overall assemblage structure would occur at the larger spatial scale, hundreds of kilometers between bays. We also expect a strong relationship between changes in assemblage structure and the PAH contamination gradient.

2. Material and methods

2.1. Study area

Guanabara Bay (GB; Fig. 1b), located in the Rio de Janeiro metropolitan region (22°80′S, 43°15′W), is a large coastal bay (384 km^2) with great economic, social, cultural and ecological relevance (Soares-Gomes et al., 2016). The bay harbors the second largest industrial complex of Brazil, an oil refinery and many oil terminals, two commercial ports and is surrounded by the largest coastal urban settlement in the country, with > 11 million inhabitants. Consequently, the bay receives a large load of inorganic and organic contaminants delivery by fluvial input, atmospheric deposition and urban runoff, making it one of the most impacted estuaries in the Brazilian coast (e.g., Kjerfve et al., 1997; Carreira et al., 2004; Baptista Neto et al., 2006; Wagener et al., 2012; Fistarol et al., 2015). High concentrations of hydrocarbons are usually found in the northwest portion of GB, close to the port region and oil refinery (Meniconi et al., 2002; Wagener et al., 2012). On the other hand, the northeast portion, which includes the environmental protection area, has significantly lower concentrations, being one of the few locations of GB where severe contamination by hydrocarbons is not observed (Mauad et al., 2015).

Laranjeiras Bay (LB; Fig. 1a), located in the north-south axis of the Paranaguá Estuarine Complex (612 km^2) in Paraná state (25°40′S, 48°37′W), is a semi-closed bay (240 km^2) bordered by extensive tidal flats, mangroves and saltmarshes. This region presents a great diversity

of preserved habitats (Sardi et al., 2016) and includes 20% of the total remnant Brazilian Atlantic forest (Abreu-Mota et al., 2014). The most important economic activities are small-scale fisheries, incipient aquaculture, and tourism (Martins et al., 2012). Despite the existence of potential impact sources in nearby areas, such as oil terminal operations and sewage discharges from Paranaguá city, LB is not contaminated by petroleum derivates and can still be classified as pristine (Martins et al., 2012).

2.2. Sampling design and field procedures

The sampling design included four scales of spatial variation: Bays (100′s km), Sectors (km), Locations (100′s m) and Sites (10′s m). In each bay, the inner (S1), intermediate (S2) and external (S3) sectors were defined along the salinity-energy gradient (Fig. 1a,b). Within each sector, three locations (200 to 500 m apart from each other) were randomly selected. In each location, two sites tens of meters apart were chosen, and within these, four sediment samples were collected using a Van Veen grabber. Each sediment sample was sieved through a 0.5 mm mesh and fixed in 8% formalin; animals were counted and identified to the lowest taxonomic level with a stereo-microscope.

An additional sediment sample was taken from each site to determine the concentration of hydrocarbons (AHs and PAHs) and sediment granulometry. For hydrocarbon analysis, the top 2 cm of surface sediment was collected with a spoon and placed in pre-cleaned aluminum foil and stored at − 20 °C. The material was freeze-dried, carefully homogenized with a mortar, and stored in clean glass bottles at room temperature prior to hydrocarbon analysis. For granulometric analysis, the top 2 cm of surface sediment was collected with a spoon, placed in plastic bags and stored at − 20 °C until processing.

2.3. Laboratory procedures

The methodology to hydrocarbons was based in the methods EPA-8015B and EPA-8270D, to aliphatic hydrocarbons and polycyclic aromatics, respectively. Between 5 and 10 g of lyophilized sediment were extracted in an ASE (accelerated solvent extraction) equipment after addition of 2500 ng of n-$C_{16}D_{34}$, 2500 ng of n-$C_{30}D_{62}$ (for aliphatic hydrocarbons – AHs) and 100 ng of p-terfenil-D_{14} (for PAHs) as surrogates for validation of the method.

After the extraction, the aliphatic and aromatic fractions were separated from the crude extract through liquid chromatography on open column using 10 g of 5% deactivate silica, 7 g of 2% deactivate alumina and 1 g of sodium sulfate. Fractions were isolated with 50 mL of hexane (F1 – AHs) followed by 100 mL of hexane:dichloromethane (1:1) (F2 – PAHs).

The resulting extracts of each fraction were concentrated until 1 mL in a Turbovap® and received the internal quantification standards (F1: 2500 ng.mL^{-1} of n-C_{24} deuterated; F2: 100 ng.mL^{-1} of naphtalene-D_8, 100 ng.mL^{-1} acenaphtene-D_{10}, 100 ng.mL^{-1} phenanthrene-D_{10}, 100 ng.mL^{-1} chrysene-D_{12} e 100 ng.mL^{-1} perylene-D_{12}).

The aliphatic hydrocarbons were quantified using a gas chromatograph equipped with flame ionization detector (GC-FID-HO6890) and the PAHs with gas chromatograph fitted to a mass spectrometer (GC–MS). The detection and quantification limits were 2 to 9 ng g^{-1}, respectively, to individual aliphatic hydrocarbons and 0.02 to 0.07 ng g^{-1}, respectively, for individuals PAHs, being considered as 0,00 to statistical analysis. The analytical quality control was based in the determination of blanks through all the process and the calculation of the recovery percentage of the surrogates. Individual n-alkanes in low amounts were found in the blanks and these values were subtracted from the sample readings. The recovery of the surrogates between 40 and 120% were considered acceptable.

Among the aliphatic hydrocarbons were quantified the individuals n-alkanes (n-C_{12} to n-C_{40}), isoprenoids (pristane and phytane) and the unresolved complex mixture (UCM). A total of 37 PAHs were analyzed, including the 16 priority compounds of the US Environmental Protection Agency (EPA) (Buchman, 2008), besides the alkylated compounds such 1-methyl- and 2-methyl-naphtalene (considered as C1-naphatalenes), C2 and C4-naphtalenes, C1 to C3-fluorenes, C1 to C4-phenanthrenes, C1 to C3-dibenzothiophenes, C1 and C2-pyrenes and C1 and C2-chrysenes.

Percentages of organic matter and biodetritic carbonate were obtained through the method described by Gross (1971). The organic matter was measured gravimetrically by sediment weight loss after oxidation by a solution of hydrogen peroxide (H_2O_2) at 30%, whereas calcium carbonates were measured similarly in another aliquot of sediment after treatment with HCl 10%. The carbonate and organic matter-free sediment was analyzed in a Bluewave S5400 laser granulometer, for definition of the mean grain size, standard deviation (i.e. degree of selection), asymmetry and kurtosis.

2.4. Assessment of the origin of hydrocarbons

The origin of the aliphatic hydrocarbons was analyzed through the following indexes:

- Carbon Preference Index – CPI (Wang et al., 1999);
- Terrestrial Aquatic Ratio – TAR (Bourbonniere and Meyers, 1996);
- Average Chain Length – ACL (Belligotti et al., 2007).

The source assignments of the PAHs were based on diagnostic ratios that involve compounds with the same molecular weight, but with different thermodynamic stabilities (Yunker et al., 2002), including:

- Fluoranthene/Fluoranthene + Pyrene - [Fl/(Fl + Py)];
- - Indene[1,2,3-c,d]Pyrene/Indene[1,2,3-c,d]Pyrene + Benzo(g,h,i) Perylene - [I-Py/(I-Py + BghiPe)];
- Benzo(a)Anthracene/Benzo(a)Anthracene + Crysene - [BaA/(BaA + Cr)];
- Anthracene/Anthracene + Phenantrene - [An/(An + Ph)].

2.5. Statistical analysis

The mean grain size, organic matter and calcium carbonate contents and aliphatic and polycyclic aromatic hydrocarbons concentrations were separately analyzed by a three-factor analysis of variance with the following model: Bays (fixed, 2 levels: GB, LB), Sector (fixed, 3 levels crossed with Bays: S1, inner; S2, intermediate; S3, external) and Locations (random, 3 levels nested to the interaction between Bay and Sector). A posteriori comparisons in significant terms of interest (α = 0,05) were conducted by the Student-Newman-Keuls procedure (SNK). The homogeneity of variances was verified using Cochran's test and the data was transformed when necessary (Underwood, 1997).

Differences in the total number of individuals, number of taxa, Shannon-Weaver diversity index and the density of the six numerically dominant species were individually tested by a four-factor analysis of variance (ANOVA) with the following linear model: Bay (fixed, 2 levels: GB and LB), Sector (fixed, 3 levels crossed with Bays: S1, internal; S2, intermediate; S3, external), Locations (random, 3 levels nested in the interaction between Bay and Sector) and Sites (random, 2 levels nested in Location). Components of variation (i.e. magnitude of effects) were calculated for all terms of the model. The homogeneity of the variances was verified by the Cochran's test and the data was transformed when necessary (Underwood, 1997).

Differences among macrofaunal assemblages were tested by a permutational multivariate analysis of variance (Anderson, 2001) using the same linear model from the univariate analysis. A non-metric multidimensional scaling analysis (nMDS) was used to visualize patterns of assemblage structure between Bays and Sectors. Similarity percentages analysis (SIMPER) was applied to identify the contribution of typically abundant macrofaunal taxa to the total dissimilarity in

assemblages between bays. All multivariate analyses were performed using the Bray-Curtis similarity index on ln(x + 1) transformed data.

A canonical analysis of principal coordinates (CAP, Anderson, 2008) was used to model the variations in the structure of the macrofaunal assemblages along the gradient of contamination by polycyclic aromatic hydrocarbons. For this, the individual concentrations of the 16 PAHs in sediment samples were combined into a single gradient of contamination by the principal component analysis (PCA). This 16 were chosen for being the priority PAHs used in studies of environmental pollution. The scores from the first principal component (PC1) were used as a proxy variable of the contamination by PAHs in the CAP. The analysis was based in the coefficients of the Bray-Curtis similarity with transformed data to ln(x + 1). The CAP is specifically formulated to identify an axis (i. e. gradient) in the multivariate cloud of biotic data that has the stronger relation with the abiotic variable of interest (Anderson et al., 2008), even in the presence of other potential sources of variability responsible for differences in assemblage structure. Excessive parameterization of the model was controlled by the choice of 10 main coordinate axes (PCO, m = 10).

PERMANOVA and CAP were performed using PERMANOVA + add-on package for PRIMER v6 (Clarke and Gorley, 2006; Anderson et al., 2008). All other analyzes and graphs were produced in the R language (R Core Team, 2013), using the packages GAD (Sandrini-Neto and Camargo, 2012), vegan (Oksanen et al., 2012) and sciplot (Morales, 2012).

3. Results

3.1. Sediment

The mean grain size, organic matter and biodetritic carbonate content in the sediment are presented as supplementary material (Table S1). Mean grain size in GB varied from coarse silt (60 μm) to medium sand (500 μm), while in LB sediments varied from very fine sand (125 μm) to medium sand. Nonetheless, no significant differences were detected in the mean grain diameter (Table 1a) and sediments were predominantly composed by fine sand (125–250 μm) in both bays (Fig. 2).

Variations in the organic matter and the biodetritic carbonate contents were caused by the combination between bays and sectors (significant Ba × Se interactions; Table 1b,c). Planned comparisons between bays for each level of sector revealed that organic matter and carbonate contents were significantly higher in GB, except in the external sector, where no differences were observed between the bays (SNK tests; Table 1b,c).

For comparisons among sectors within each bay, percentages of organic matter and carbonates were significantly higher in the internal (S1) and intermediate (S2) sectors of GB, which did not differ between each other (SNK tests; Table 1b,c). In LB, organic matter content was significantly higher in the internal sector, not differing between external and intermediate sectors (SNK tests, Table 1b). There were no significant differences in the percentage of biodetritic carbonate between the sectors of LB (SNK tests; Table 1c).

3.2. Aliphatic hydrocarbons

The total aliphatic concentration, unresolved complex mixture (UCM) and the indices for assessing the origin of the compounds are discriminated in the Table S1. The total aliphatic (ΣAH) varied from 18.66 to 516.8 μg g^{-1} of dry sediment (147.5 ± 127.7; mean ± standard deviation) in GB and from 0.54 to 16.52 μg.g^{-1} (4.5 ± 4.8; mean ± standard deviation) in LB (Fig. 2). Concentrations of total aliphatic hydrocarbons were significantly higher in GB than LB, despite a significant Ba × Se interaction (SNK tests; Table 1d; Fig. 2). Comparisons among sectors for each bay showed that concentrations of total aliphatic hydrocarbons in GB were significantly higher in the intermediate sector and lower in the external sector (SNK tests. Table 1d). In LB, the ΣHA was significantly higher in the inner sector, and did not differ between the intermediate and external sectors (SNK test; Table 1d; Fig. 2).

The unresolved complex mixture (UCM) reached values > 90% of total aliphatics at all sites of the intermediate and external sectors of GB. The presence of UCM was only recorded in 10 samples (inner sector) of LB, with an average of 23% of total aliphatics (Table S1).

The ratio between UCM and the resolved aliphatic (RA, compounds that could be identified through chromatography) varied from 0.4 to 14.4, indicating the presence of residues of petroleum hydrocarbons in all sites of the intermediate and external sectors of GB. The ratio UCM:RA varied from 0.0 to 1.7 in LB, evidencing the presence of natural hydrocarbons.

The carbon preferential index (CPI) varied from 1.6 to 4.0 in GB and from 2.0 to 8.5 in LB, indicating aliphatic hydrocarbons of petrogenic origin in the inner sector of GB and hydrocarbons from natural sources, predominantly terrestrial, in practically all sites of LB.

The terrestrial aquatic ratio (TAR) index used to distinguish the marine or terrestrial origin of the natural hydrocarbons indicated that all sites, in both bays, are composed predominantly from terrestrial sources, especially higher plants such as those found in mangroves forests.

The average chain length (ACL) was 30 ± 0.2 (mean ± standard deviation) in GB and 29.5 ± 0.2 (mean ± standard deviation) in LB. These are values similar to those found in mangrove leaves of warm regions (Belligotti et al., 2007), indicating the presence of terrestrial organic matter from the mangroves in both regions.

3.3. Aromatic hydrocarbons

The total polycyclic aromatic hydrocarbons concentrations (ΣPAHs), 16 priority PAHs, higher and lower molecular weight PAHs, percentage of perylene and ratios that investigates the origin of the compounds are discriminated in Table S1.

The total PAHs varied from 101.3 to 4148 ng.g^{-1} of dry sediment (910.4 ± 1030; mean ± standard deviation) in GB and from 0.52 to 8.27 ng.g^{-1} (4.02 ± 2.58; mean ± standard deviation) in LB. The analysis of variance identified significantly higher concentration of the Σ16 PAHs in GB compared to LB, independently of the sampled sector (SNK tests; Table 1e; Fig. 2). A posteriori tests comparing sectors within each bay showed that concentrations of the Σ16 PAHs were significantly higher in the intermediate sector of GB, and did not differ between the internal and external sectors (SNK tests; Table 1e). There were no significant differences among sectors of LB (SNK tests; Table 1e).

Almost all samples of GB showed predominance of high molecular weight PAHs (HMW – 4 to 6 rings), indicating pyrogenic sources that may include vehicular and industrial emissions (Wagener et al., 2012). Low molecular weight PAHs (LMW - 2 and 3 rings) were predominant in LB, suggesting the presence of petroleum, although in very low concentrations (Yunker et al., 2002).

For the limits TEL (*Threshold Effect Level*), ERL (*Effect range-low*) and PEL (*Probable Effect Level*) – defined by the US environmental protection agency (EPA) for the 16 priority PAHs for studies of environmental quality (Buchman, 2008) – GB showed higher values than the TEL limit for the total PAHs and for some individual compounds (acenaphtene, benz(a)anthracene and chrysene) in all sites of the intermediate sector. Moreover, in all sites of this sector, the values were higher than the ERL limit for anthracene. None of the compounds reached the adverse effects levels in LB.

The percentage of perylene to its isomers was lower than 10% in most of the GB sites and higher in all sites of LB, indicating perylene from anthropic and natural sources, respectively (Jiang et al., 2009).

The ratios between the isomers with the same molecular mass showed that GB is under influence of PAHs from pyrolytic origin

Table 1
Summary of analysis of variance ($n = 2$ replicate sites) for mean grain size, organic matter content, calcium carbonate content, total aliphatic hydrocarbons and 16 polycyclic aromatic hydrocarbons. For SNK pair-wise a posteriori comparisons: GB = Guanabara Bay; LB = Laranjeiras Bay. " > " indicates $p < 0.05$ and "=" indicates $p > 0.05$. Significant terms of interest ($\alpha = 0.05$) are highlighted in bold. Data transformed to [1]square root, [2]ln(x + 1) and [3]arc-sin before analysis.

		(a) Mean grain size		(b) Organic matter[3]		(c) Calcium carbonate[3]	
Source	df	MS	*F*	MS	*F*	MS	*F*
Bay = Ba	1	75.17	0.022	0.209	34.147***	0.296	34.242***
Sector = Se	2	2864.07	0.849	0.170	27.674***	0.096	11.099**
Ba × Se	2	3019.52	0.895	0.048	**7.905****	0.049	**5.626***
Location(Ba × Se)	12	3373.22	0.699	0.006	2.297	0.009	0.961
Residual	18	4828.88		0.003		0.009	
SNK tests				Among levels of **Se**		Among levels of **Se**	
				GB: S3 < S1 = S2		GB: S3 < S1 = S2	
				LB: S3 = S2 < S1		LB: S3 = S2 = S1	
				Between levels of **Ba**		Between levels of **Ba**	
				S1: LB < GB		S1: LB < GB	
				S2: LB < GB		S2: LB < GB	
				S3: LB = GB		S3: LB = GB	

		(d) ΣAH[2]		(e) Σ16 PAH[1]	
Source	df	MS	*F*	MS	*F*
Bay = Ba	1	94.000	417.871***	5383.935	68.672***
Sector = Se	2	3.359	14.931***	549.001	7.003***
Ba × Se	2	5.566	**24.745*****	612.489	**7.812****
Location(Ba × Se)	12	0.225	0.759	78.400	2.174
Residual	18	0.296		36.062	
SNK tests		Among levels of **Se**		Among levels of **Se**	
		GB: S3 < S1 < S2		GB: S1 = S3 < S2	
		LB: S3 = S2 < S1		LB: S1 = S2 = S3	
		Between levels of **Ba**		Between levels of **Ba**	
		S1: LB < GB		S1: LB < GB	
		S2: LB < GB		S2: LB < GB	
		S3: LB < GB		S3: LB < GB	

Significance codes: *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

(combustion of fossil fuel, coal, petroleum) (Fig. 3a,b), while LB is subjected by multiples sources, both pyrolytic and petrogenic (Fig. 3c,d).

3.4. Benthic macrofauna

A total of 75,538 individuals from 27 taxa were collected in GB. The gastropod *Heleobia australis* was the most abundant organism, comprehending > 91% of total abundance in this estuary. A total of 7558 individuals belonging to 55 taxa were counted in LB. Polychaetes were the most dominant and diverse group, with 28 taxa. The polychaetes *Aricidea* sp. and *Lumbricalus januarii* contributed with 18% and 6% of the total abundance, respectively. Oligochaetes from the subfamily Tubificidae also occurred in great numbers, representing 21% of the total macrofauna in this bay.

The total number of individuals was significantly higher in GB (Table 2a; Fig. 4), but number of taxa and the Shannon-Weaver diversity index revealed assemblages less diverse and with high dominance when compared to LB (Table 2b,c; Fig. 4). The total number of individuals, number of taxa, Shannon-Weaver diversity and the density of the six numerically dominant taxa showed significant differences on the smallest spatial scales adopted (i.e. sites and locations) (Table 2). However, estimates of the components of variation (i.e. magnitude of effects) calculated for each term of the analysis showed that larger scales contribute significantly to the total variability of most benthic descriptors and taxa, particularly for the comparison between bays (Table 2).

Differences between GB and LB bays accounted for about 42% of total number of individuals´ variance (Table 2a; Fig. 4). The number of taxa and Shannon index, although significantly different between locations, were largely variable between bays, which contributed to 37 and 53% of the total variability, respectively (Table 2b,c; Fig. 4). Most of the patterns described for the total number of individuals is due to the contribution of the gastropod *H. australis*, which exhibited significantly higher densities in GB (Fig. 4). The analysis of the components of variation revealed that the difference between the bays was responsible for nearly 66% of the total variability in *H. australis* density (Table 2d; Fig. 4).

The occurrence of the polychaetes L. *januarii* and *M. papillicornis* was restricted to the intermediate and external sectors of LB (Fig. 4). Components of variation calculated for both species (Table 2f,h) revealed great contribution of bays, but also showed that the interaction Ba × Se is equally important in defining the distribution patterns of both species. Densities of the spionid polychaete *Streblospio benedicti* were strongly variable among sectors, which accounted for 30% of total variation (Table 2e). This pattern was primarily related to peaks in the locations 2 and 3 in the external sector of GB (Fig. 4). Generally, distribution patterns of oligochaetes and the polychaete *Aricidea* sp. were rather similar. Components of variation for bays, sectors and their interaction accounted for > 60% of total variance of these taxa (Table 2g,i; Fig. 4).

Significant differences in the structure of macrofaunal assemblages were detected in all sources of variation considered in the model (Table 3). Despite the significant differences at smallest spatial scales indicated by PERMANOVA, the analysis of the components of variation (Table 3) revealed that differences between bays contributed with 47% of the total dissimilarity of assemblages. The combination between Bays and Sectors, represented by the interaction Ba × Se, was the second most important component, explaining nearly 18% of macrofauna dissimilarity (Table 3).

This pattern is clearly illustrated by nMDS plot (Fig. 5), which revealed an evident distinction between GB and LB, as well as differences

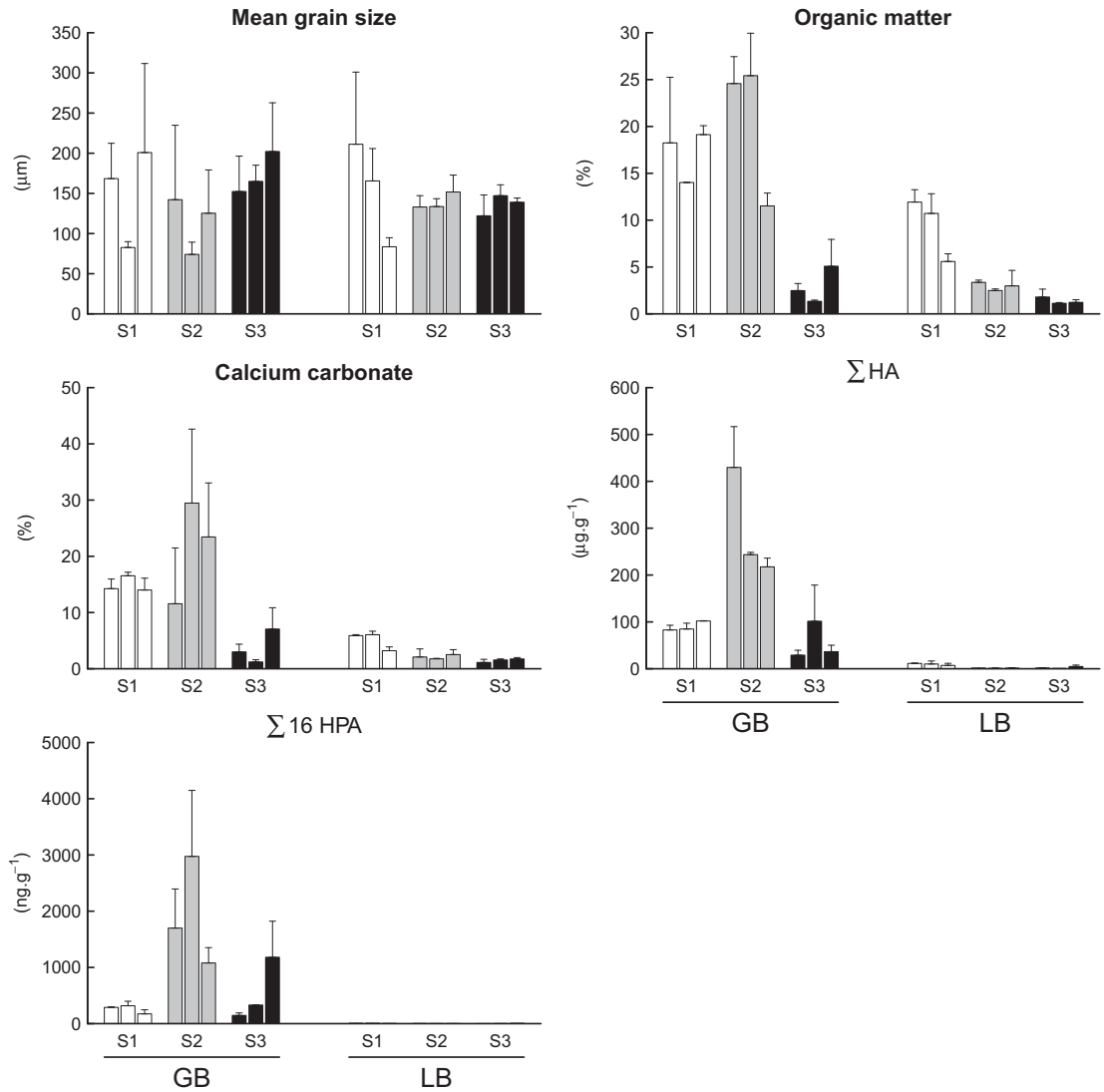

Fig. 2. Mean (SE) grain size, organic matter content, calcium carbonate content, total aliphatic hydrocarbons (ΣHA) and 16 polycyclic aromatic hydrocarbons (Σ16 HPA) between GB and LB. Each bar is a location within each sector and is the mean of 2 sites. S1 = inner sector; S2 = intermediate sector; S3 = external sector.

among sectors within each bay. In GB, assemblage structure in the inner sector is different from those at intermediate and external sectors. In LB, assemblage structure in the external sector differs from those at the intermediate and external sectors (Fig. 5). SIMPER analysis showed that *H. australis* alone accounted for 82% of the total similarity within GB. Oligochaetes and the polychaetes *Aricidea* sp., *S. benedicti* and L. *januarii* contributed with 47% of the similarity within LB.

3.5. Relationship between macrofaunal assemblages and the PAH contamination gradient

The first component of PCA (PC1) explained 90% of Σ16 PAHs variability in sediment samples, and can clearly serve as a useful proxy variable for the overall gradient of hydrocarbon contamination. A strong relationship between changes in macrofaunal assemblage structure and the PAH contamination gradient was observed in CAP analysis, with a canonical correlation of $\delta = 0.95$ using $m = 10$ principal coordinate axis (Fig. 6). The most contaminated sites were from GB, particularly in the intermediate sector.

4. Discussion

Despite some small-scale variability in macrofauna taxa, most of the variation occurred at the largest spatial scale. Therefore, our hypothesis that the largest proportion of total variation in macrofauna density, diversity and overall assemblage structure would occur at this scale of variability (i.e. hundreds of kilometers between bays) was not rejected. We have also showed that the concentrations of aliphatic and polycyclic aromatic hydrocarbons were significantly higher in GB than LB. Finally, we demonstrated a strong and consistent relationship between the structure of macrofauna and the gradient of contamination by PAHs, showing that this class of hydrocarbons is one of the main structuring factors of soft-bottom assemblages in both bays.

Aliphatic hydrocarbons in sediments of both GB and LB were predominantly derived from terrestrial sources, mainly mangrove forests surrounding the bays. Diagnostic ratios of source assignments of PAHs showed a predominance of pyrolytic sources of these compounds in GB sediments, including combustion of fossil fuels and, to a lesser extent, plant biomass (Meniconi et al., 2002; Wagener et al., 2012). The specificity of common diagnostic ratios may be, however, masked by the high level of degradation of organic matter in the bay (Massone et al., 2013). Additional evidence of the limitation of diagnostic ratios in source assignments of PAHs is shown by the pattern of the alkylated series of the samples, with the distribution showing that the hydrocarbons with lower number of alkylation (lighter) are in lower concentrations than those with higher alkylated groups (Page et al., 1995).

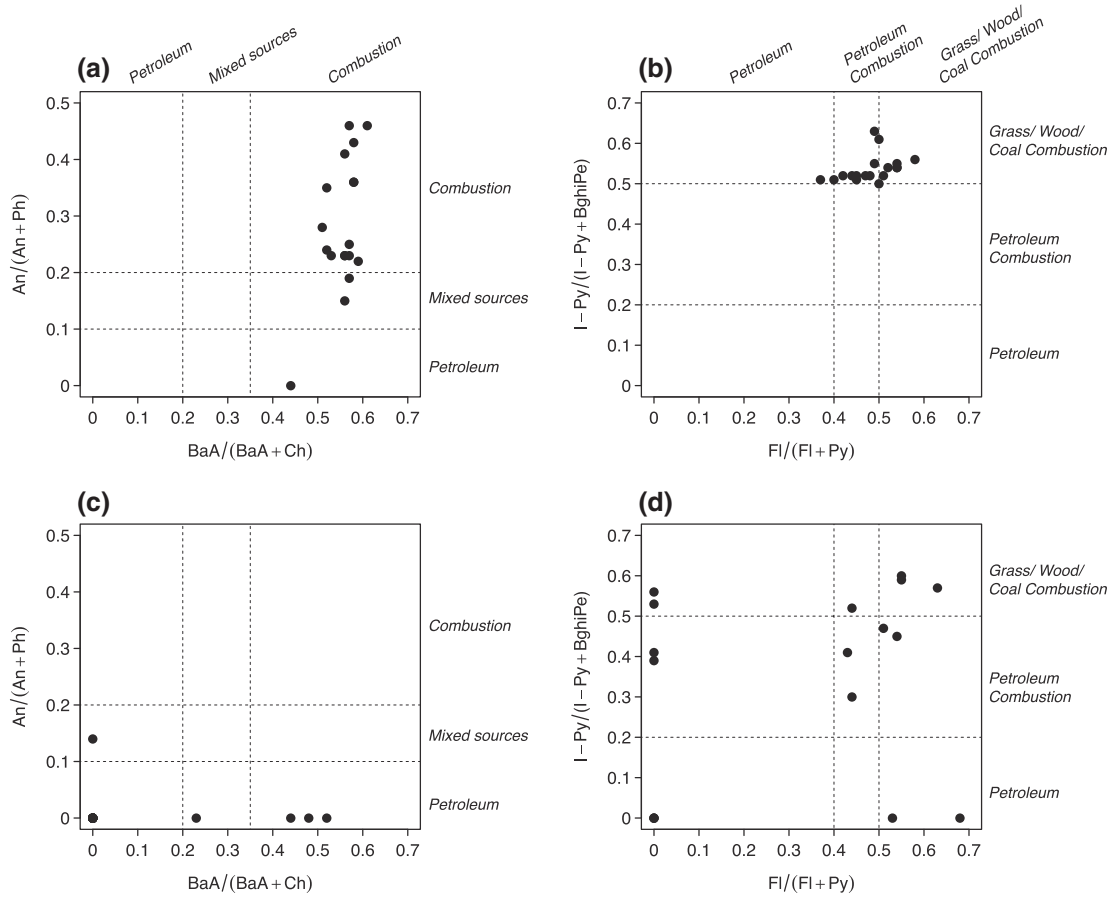

Fig. 3. PAH cross plots for the ratios of (a) BaA/(BaA + Ch) versus An/(An + Fe) and (b) Fl/(Fl + Py) versus I-Py/(I-Py + BghiPe) for sediments of GB; and (c) BaA/(BaA + Ch) versus An/(An + Fe) and (d) Fl/(Fl + Py) versus I-Py/(I-Py + BghiPe) for sediments of LB.

The use of sampling designs with different scales of variation is a widespread approach to quantify distribution of benthic macrofauna (Chapman et al., 2010; Veiga et al., 2016, Ysebaert and Herman, 2002), but it also proved to be an appropriate tool to describe spatial patterns of hydrocarbons, especially in LB. In our study, total aliphatics reached concentrations ca. 50 times higher than those found by Martins et al. (2012) in the inner sector of the bay. This difference is probably due to small-scale variation of total aliphatics within the inner sector of LB, since the concentration found by Martins et al. (2012) was based on a single sample and extrapolated to the entire area. Concentrations based on punctual samplings are potentially confounded, thus emphasizing the need of replication at a hierarchy of spatial scales (see Morrisey et al., 1994).

The clear distinction in the structure of the macrobenthic assemblages between the two bays analyzed was evident in the nMDS ordination plot and by the components of variation in PERMANOVA. This difference is possibly related to distinct structuring processes operating in each bay, which are conditioned by abiotic factors, such as salinity, sedimentary characteristics and depth; biotic factors, such as food availability and competition; and by anthropic influences, such as discharge of effluents, oil spills, industrial activities, among others (Barros et al., 2008; Venturini et al., 2008). The incorporation of multiple scales of spatial variation is important to compare estuaries that are apart from one another, since the structure of macrobenthic assemblages may vary within each system due to multiple sources of heterogeneity (Giménez et al., 2014).

In GB, the total number of individuals decreased from the inner to the external sectors, while the number of taxa and diversity increased, in accordance to the patterns described by Santi and Tavares (2009) and Neves et al. (2012). The general pattern of low diversity and high abundance in GB is mainly due to the contribution of the gastropod *Heleobia australis*, which accounted for 91% of total macrofauna density and was particularly abundant in the inner sector of the GB. This sector has a high level of organic matter and is constituted by anoxic-hypoxic muddy sediments (Mendes et al., 2006) that result in a major impoverishment of the benthic fauna, but may favor high dominance of opportunistic species. *H. australis* is an opportunistic surface deposit feeder often found in degraded sites (Echeverría et al., 2010; Carcedo and Fiori, 2012).

A similar pattern of high dominance by *H. australis* has been previously described in other estuarine systems of South America, particularly in Montevideo Bay, Uruguay (Venturini et al., 2004; Muniz et al., 2011; Muniz and Venturini, 2015). Similarly to GB, sediments of the inner part of Montevideo Bay contain high loads of organic matter and large amounts of contaminants, such as metals and petroleum hydrocarbons from multiple diffuse sources, including sewage, marine traffic and a petroleum refinery (Venturini et al., 2004).

Streblospio benedicti was the dominant polychaete in GB, where it occurred in the external and intermediate sectors (the most contaminated by hydrocarbons), but was practically absent in the inner sector, where no hydrocarbon contamination is observed despite the high organic load. *S. benedicti* is an opportunistic spionid polychaete frequently found in environments that are under stress (Dauer et al., 2003). However, this species is not as tolerant to drastic reductions of dissolved oxygen, observed in the inner sector of GB, as to high levels of contamination (Reish, 1979; Llansó, 1991; Mendes and Soares-Gomes, 2013).

In LB, the total number of taxa and diversity were significantly higher compared to GB, with dominance of the paraonid polychaete *Aricidea* sp. and a Tubificinae oligochaete, that altogether accounted for

Table 2
Summary of analysis of variance and percentage components of variation (n = 4 replicate cores) for total density of macrofauna, total number of taxa, Shannon-Weaver diversity index and densities of numerically dominant taxa. Data transformed to [1]square root, [2]fourth root and [3]ln(x + 1) before analysis.

		(a) Total individuals[3]			(b) Total taxa[1]			(c) Shannon-Weaver[3]		
Source	df	MS	*F*	%v.c.	MS	*F*	%v.c.	MS	*F*	%v.c.
Bay = Ba	1	209.77	138.10***	42.0	162.09	143.24***	37.3	28.60	220.92***	52.7
Sector = Se	2	11.99	7.89**	11.5	70.61	62.40***	30.0	3.24	25.04***	21.3
Ba × Se	2	14.71	9.69**	18.3	7.63	6.74*	13.0	0.28	2.15	6.6
Location(Ba × Se)	12	1.52	1.63	6.7	1.13	7.44***	8.7	0.13	15.78***	10.3
Site(Lo(Ba × Se))	18	0.93	4.77***	10.6	0.15	1.21	2.0	0.01	0.70	–
Residual	108	0.20		10.9	0.13		8.9	0.01		9.1

		(d) *Heleobia australis*[3]			(e) *Streblospio benedicti*[2]			(f) *Lumbricalus januarii*[1]		
Source	df	MS	*F*	%v.c.	MS	*F*	%v.c.	MS	*F*	%v.c.
Bay = Ba	1	1304.64	307.84***	65.7	16.92	16.77**	15.4	141.44	45.25***	28.9
Sector = Se	2	12.18	2.87	6.3	41.81	41.42***	30.2	40.16	12.85**	18.3
Ba × Se	2	6.02	1.42	4.2	12.99	12.87**	23.2	40.16	12.85**	25.9
Location(Ba × Se)	12	4.24	5.24***	10.1	1.01	2.61*	9.1	3.13	5.77***	11.8
Site(Lo(Ba × Se))	18	0.81	3.08***	5.7	0.39	1.75*	6.7	0.54	3.04***	6.3
Residual	108	0.26		7.9	0.22		15.4	0.18		8.8

		(g) *Aricidea* sp.[1]			(h) *Magelona papillicornis*[3]			(i) Oligochaeta[3]		
Source	df	MS	*F*	%v.c.	MS	*F*	%v.c.	MS	*F*	%v.c.
Bay = Ba	1	201.15	12.86**	20.5	16.49	14.80**	23.2	70.27	77.87***	21.2
Sector = Se	2	135.21	8.65**	20.2	4.66	4.18*	13.6	67.48	74.77***	25.5
Ba × Se	2	94.96	6.07*	23.3	4.66	4.18*	19.3	39.93	44.25***	27.6
Location(Ba × Se)	12	15.64	4.25**	15.6	1.11	3.63**	15.9	0.90	0.62	–
Site(Lo(Ba × Se))	18	3.68	7.50***	11.4	0.31	2.67***	11.0	1.45	2.96***	10.6
Residual	108	0.49		9.0	0.12		17.0	0.49		15.1

Significance codes: *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

35% of the average similarity of this estuary. Distribution patterns of the polychaetes *Aricidea* sp., *Magelona papillicornis* and *Lumbricalus januarii* suggests that these taxa are very sensitive to contamination, especially the last two species that were restricted to intermediate and external sectors of LB.

Oligochaetes were practically absent in the inner and intermediate sectors of GB, despite their tolerance to pollution and hypoxia, since they are capable of considerably lower their respiration rate to be able to survive in such conditions (Chapman, 2001; Giere, 2006). However, development and reproduction rates of oligochaetes are heavily affected by high organic load and contamination (Giere and Pfannkuche, 1982), which may explain the higher abundances observed in all sectors of LB.

The definition of a gradient of contamination by the 16 PAHs obtained from the first component of the PCA was particularly useful, especially if compared to common used descriptor such as the total PAH concentration. The first axis of the PCA explained 90% of the total variability of these compounds in an integrated way and revealed a clear gradient with high concentration in the intermediate sector of GB and an overall lower concentration in sediments of LB. As multivariate patterns of benthic assemblages are more sensitive to environmental changes than single indicator taxa or biotic indices (Anderson, 2008), the relationship between macrofauna and PAH gradient was verified by the canonical analysis of principal coordinates (CAP). CAP is a very useful statistical technique for the analysis of a single gradient, despite the existence of other potentially important factors structuring assemblages (Anderson et al., 2008).

Nonetheless, it is important to acknowledge that soft-bottom assemblage structure is also conditioned by other relevant sources of variation, which may be overlapped to the PAH contamination gradient. For example, high levels of faecal steroids and metals are frequently found in fine sediments heavily loaded by organic matter (Muniz et al., 2004). Sediments act as a sink for pollutants from numerous human activities, and higher concentrations are usually associated with the cohesive sediment fraction (i.e., silt and clay) (Ellis et al., 2015). Quantification of other pollutants would therefore improve the characterization of contamination in GB and LB and its relationship to changes in macrofaunal assemblages.

The model generated to characterize changes in assemblage structure with increasing PAH concentrations in sediments can be considered robust (given the high canonical correlation $\delta = 0.95$), but somehow limited due to a single sampling campaign. One of the main qualities of the CAP is its predictive characteristic, given as the ability to correctly allocate new samples along the initially described gradient of contamination. Thus, samplings of the benthic macrofauna and hydrocarbons over broader temporal scales (e.g. years) are important to integrate the responses in time and consequently improve the predictive capacity of the model. Although the proposed model does not establish causal relations between the contamination gradient and the structure of macrofaunal assemblages, it represents a significant step forward for the evaluation of PAH pollution in estuarine sediments, especially considering its predictive potential with the expansion of the temporal and spatial scales adopted.

5. Conclusion

The use of several spatial scales in a well-replicated sampling design proved to be a useful tool when assessing the influence hydrocarbon pollution on soft-bottom assemblages. Comparisons between Guanabara and Laranjeiras bays using several biotic and abiotic variables revealed that these large estuaries are very different from one another in terms of contamination levels and assemblage structure and diversity.

We isolated the PAH contamination gradient using CAP analysis to model the relationship between macrofaunal assemblages and the concentration of polycyclic aromatic hydrocarbons. The high canonical

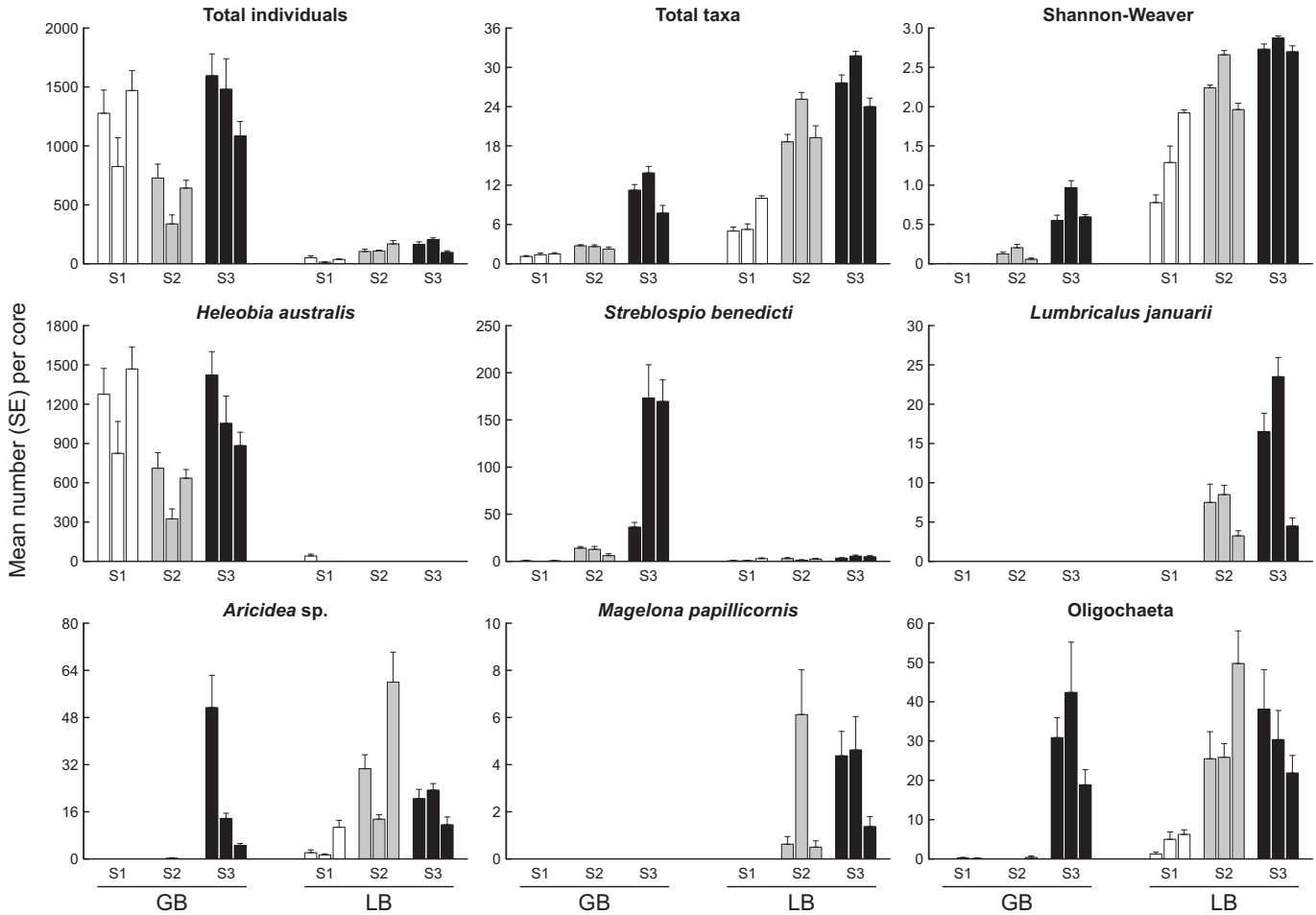

Fig. 4. Mean (SE) density of macrofauna, number of taxa, Shannon-Weaver diversity index and densities of dominant taxa between GB and LB. Each bar is a location within each sector and is the mean of 4 cores × 2 sites = 8 cores. S1 = inner sector; S2 = intermediate sector; S3 = external sector.

Table 3
Summary of PERMANOVA and percentage components of variation (9999 permutations, n = 4 replicate cores) based on Bray-Curtis dissimilarities of ln(x + 1) transformed macrofaunal densities.

Source	df	MS	*Pseudo-F*	%v.c.
Bay = Ba	1	150,094.97	50.50***	47.2
Sector = Se	2	34,057.89	11.46***	15.0
Ba × Se	2	22,101.06	7.44***	18.4
Location(Ba × Se)	12	2972.19	3.49***	6.1
Site(Lo(Ba × Se))	18	851.81	1.77***	2.1
Residual	108	482.46		11.1

Significance codes: *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

correlation coefficient indicated that this class of hydrocarbons is responsible for structuring soft-bottom assemblages in both bays. However, it should be stressed that observed patterns may be correlated to other pollutants and vary at different time scales, which were not incorporated in our study.

The following are the supplementary data related to this article.

Table S1 Environmental variables, aliphatic and polycyclic aromatic hydrocarbons and related parameters for each replicate site (R1; R2) within each location (L1; L2; L3) within each sector (S1 = inner sector; S2 = intermediate sector; S3 = external sector) of Guanabara and Laranjeiras bays. Mz, mean grain size (μm); OM, organic matter content (%); $CaCO_3$, calcium carbonate content (%); ΣAHs, total aliphatic hydrocarbons (μg.g^{-1}); UCM, Unresolved Complex Mixture (μg.g^{-1}); UCM/RA, Unresolved Complex Mixture and Resolved Aliphatic ratio; CPI, Carbon Preference Index; TAR, Terrestrial Aquatic Ratio; ACL, Average Chain Length; ΣPAHs, total polycyclic aromatic hydrocarbons (ng.g^{-1}); Σ16PAHs, 16 priority polycyclic aromatic hydrocarbons (ng.g^{-1}); LMW, low molecular weight (2–3 rings) PAHs (ng.g^{-1}); HMW, high molecular weight (4–6 rings) PAHs (ng.g^{-1}); Pe, Perylene

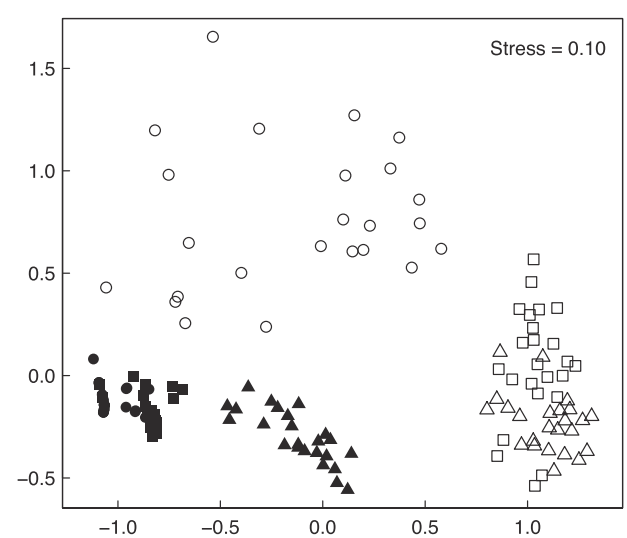

Fig. 5. Non-metric multidimensional scaling (nMDS) based on Bray-Curtis dissimilarities of ln(x + 1) transformed data comparing macrofaunal assemblages between GB (Sector 1 = ●; Sector 2 = ■; Sector 3 = ▲) and LB (Sector 1 = ○; Sector 2 = □; Sector 3 = Δ).

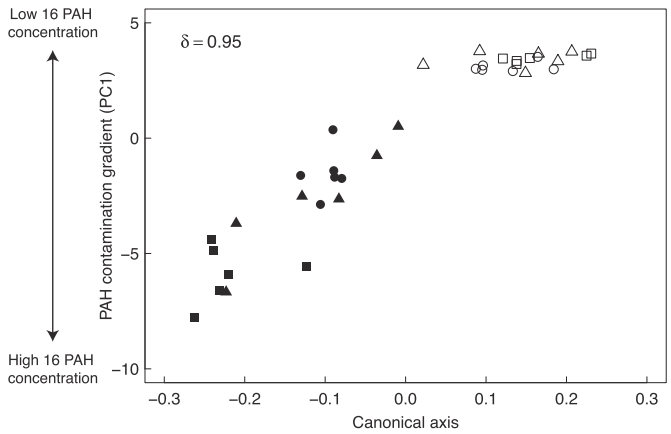

Fig. 6. Canonical analysis of principal coordinates (CAP) based on Bray-Curtis dissimilarities of ln(x + 1) transformed data relating the structure of macrofauna to the PAH contamination gradient in GB (Sector 1 = ●; Sector 2 = ■; Sector 3 = ▲) and LB (Setor 1 = ○; Setor 2 = □; Setor 3 = Δ). Each point is an average of 4 replicated cores.

content (%); Fl/(Fl + Py), fluoranthene/fluoranthene + pyrene isomer pair ratio; I-Py/(I-Py + BghiPe), Indene[1,2,3-c,d]Pyrene/Indene [1,2,3-c,d]Pyrene + Benzo(g,h,i)Perylene; BaA/(BaA + Cr), Benzo (a)Anthracene/Benzo(a)Anthracene + Crysene; An/(An + Ph), Anthr acene/Anthracene + Phenantrene.

Acknowledgements

We are grateful to Fernanda Souza and Verônica Oliveira for their invaluable help with macrofauna identification.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.marpolbul.2017.07.074.

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